Problem A. Duff and Meat

Time Limit 1000 ms
Mem Limit 262144 kB

Duff is addicted to meat! Malek wants to keep her happy for n days. In order to be happy in i-th day, she needs to eat exactly a_i kilograms of meat.

There is a big shop uptown and Malek wants to buy meat for her from there. In i-th day, they sell meat for p_i dollars per kilogram. Malek knows all numbers $a_1, ..., a_n$ and $p_1, ..., p_n$. In each day, he can buy arbitrary amount of meat, also he can keep some meat he has for the future.

Malek is a little tired from cooking meat, so he asked for your help. Help him to minimize the total money he spends to keep Duff happy for n days.

Input

The first line of input contains integer n ($1 \le n \le 10^5$), the number of days.

In the next n lines, i-th line contains two integers a_i and p_i ($1 \le a_i, p_i \le 100$), the amount of meat Duff needs and the cost of meat in that day.

Output

Print the minimum money needed to keep Duff happy for n days, in one line.

Examples

Input	Output
3 1 3 2 2 3 1	10

Input	Output
3 1 3 2 1 3 2	8

Note

In the first sample case: An optimal way would be to buy 1 kg on the first day, 2 kg on the second day and 3 kg on the third day.

In the second sample case: An optimal way would be to buy 1 kg on the first day and 5 kg (needed meat for the second and third day) on the second day.

Problem B. Alternating Subsequence

Time Limit 1000 ms
Mem Limit 262144 kB

Recall that the sequence b is a a subsequence of the sequence a if b can be derived from a by removing zero or more elements without changing the order of the remaining elements. For example, if $a = [1, 2, 1, 3, 1, 2, 1]$, then possible subsequences are: $[1, 1, 1, 1]$, $[3]$ and $[1, 2, 1, 3, 1, 2, 1]$, but not $[3, 2, 3]$ and $[1, 1, 1, 1, 2]$.

You are given a sequence a consisting of n positive and negative elements (there is no zeros in the sequence).

Your task is to choose **maximum by size** (length) *alternating* subsequence of the given sequence (i.e. the sign of each next element is the opposite from the sign of the current element, like positive-negative-positive and so on or negative-positive-negative and so on). Among all such subsequences, you have to choose one which has the **maximum sum** of elements.

In other words, if the maximum length of *alternating* subsequence is k then your task is to find the **maximum sum** of elements of some *alternating* subsequence of length k.

You have to answer t independent test cases.

Input

The first line of the input contains one integer t ($1 \le t \le 10^4$) — the number of test cases. Then t test cases follow.

The first line of the test case contains one integer n ($1 \le n \le 2 \cdot 10^5$) — the number of elements in a. The second line of the test case contains n integers $a_1, a_2, \dots, a_n$ ($-10^9 \le a_i \le 10^9, a_i \ne 0$), where a_i is the i-th element of a.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$ ($\sum n \le 2 \cdot 10^5$).

Output

For each test case, print the answer — the **maximum sum** of the **maximum by size** (length) *alternating* subsequence of a.

Examples

Input	Output
4 5 1 2 3 -1 -2 4 -1 -2 -1 -3 10 -2 8 3 8 -4 -15 5 -2 -3 1 6 1 -1000000000 1 -1000000000 1 -1000000000	2 -1 6 -2999999997

Note

In the first test case of the example, one of the possible answers is $[1, 2, \underline{3}, \underline{-1}, -2]$.

In the second test case of the example, one of the possible answers is $[-1, -2, \underline{-1}, -3]$.

In the third test case of the example, one of the possible answers is $[\underline{-2}, 8, 3, \underline{8}, \underline{-4}, -15, \underline{5}, \underline{-2}, -3, \underline{1}]$.

In the fourth test case of the example, one of the possible answers is $[\underline{1}, \underline{-1000000000}, \underline{1}, \underline{-1000000000}, \underline{1}, \underline{-1000000000}]$.

Problem C. Books

Time Limit	2000 ms
Mem Limit	262144 kB
Input File	stdin
Output File	stdout

When Valera has got some free time, he goes to the library to read some books. Today he's got t free minutes to read. That's why Valera took n books in the library and for each book he estimated the time he is going to need to read it. Let's number the books by integers from 1 to n. Valera needs a_i minutes to read the i-th book.

Valera decided to choose an arbitrary book with number i and read the books one by one, starting from this book. In other words, he will first read book number i, then book number $i + 1$, then book number $i + 2$ and so on. He continues the process until he either runs out of the free time or finishes reading the n-th book. Valera reads each book up to the end, that is, he doesn't start reading the book if he doesn't have enough free time to finish reading it.

Print the maximum number of books Valera can read.

Input

The first line contains two integers n and t $(1 \le n \le 10^5;\ 1 \le t \le 10^9)$ — the number of books and the number of free minutes Valera's got. The second line contains a sequence of n integers $a_1, a_2, \ldots, a_n$ $(1 \le a_i \le 10^4)$, where number a_i shows the number of minutes that the boy needs to read the i-th book.

Output

Print a single integer — the maximum number of books Valera can read.

Examples

Input	Output
4 5 3 1 2 1	3

Input	Output
3 3 2 2 3	1

Problem D. Quests

Time Limit 3000 ms
Mem Limit 262144 kB

There are n quests. If you complete the i-th quest, you will gain a_i coins. You can only complete at most one quest per day. However, once you complete a quest, you cannot do the same quest again for k days. (For example, if $k = 2$ and you do quest 1 on day 1, then you cannot do it on day 2 or 3, but you can do it again on day 4.)

You are given two integers c and d. Find the maximum value of k such that you can gain at least c coins over d days. If no such k exists, output `Impossible`. If k can be arbitrarily large, output `Infinity`.

Input

The input consists of multiple test cases. The first line contains an integer t ($1 \leq t \leq 10^4$) — the number of test cases. The description of the test cases follows.

The first line of each test case contains three integers n, c, d ($2 \leq n \leq 2 \cdot 10^5$; $1 \leq c \leq 10^{16}$; $1 \leq d \leq 2 \cdot 10^5$) — the number of quests, the number of coins you need, and the number of days.

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($1 \leq a_i \leq 10^9$) — the rewards for the quests.

The sum of n over all test cases does not exceed $2 \cdot 10^5$, and the sum of d over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output one of the following.

- If no such k exists, output `Impossible`.
- If k can be arbitrarily large, output `Infinity`.
- Otherwise, output a single integer — the maximum value of k such that you can gain at least c coins over d days.

Please note, the checker is **case-sensitive**, and you should output strings exactly as they are given.

Examples

<table>
<tr><th>Input</th><th>Output</th></tr>
<tr><td><pre>6
2 5 4
1 2
2 20 10
100 10
3 100 3
7 2 6
4 20 3
4 5 6 7
4 100000000000 2022
8217734 927368 26389746 627896974
2 20 4
5 1</pre></td><td><pre>2
Infinity
Impossible
1
12
0</pre></td></tr>
</table>

Note

In the first test case, one way to earn 5 coins over 4 days with $k = 2$ is as follows:

- Day 1: do quest 2, and earn 2 coins.
- Day 2: do quest 1, and earn 1 coin.
- Day 3: do nothing.
- Day 4: do quest 2, and earn 2 coins.

In total, we earned $2 + 1 + 2 = 5$ coins.

In the second test case, we can make over 20 coins on the first day itself by doing the first quest to earn 100 coins, so the value of k can be arbitrarily large, since we never need to do another quest.

In the third test case, no matter what we do, we can't earn 100 coins over 3 days.

Problem E. Riverside Curio

Time Limit 1000 ms
Mem Limit 262144 kB

Arkady decides to observe a river for n consecutive days. The river's water level on each day is equal to some real value.

Arkady goes to the riverside each day and makes a mark on the side of the channel at the height of the water level, but if it coincides with a mark made before, no new mark is created. The water does not wash the marks away. Arkady writes down the number of marks strictly above the water level each day, on the i-th day this value is equal to m_i.

Define d_i as the number of marks strictly under the water level on the i-th day. You are to find out the minimum possible sum of d_i over all days. There are no marks on the channel before the first day.

Input

The first line contains a single positive integer n ($1 \le n \le 10^5$) — the number of days.

The second line contains n space-separated integers $m_1, m_2, \ldots, m_n$ ($0 \le m_i < i$) — the number of marks strictly above the water on each day.

Output

Output one single integer — the minimum possible sum of the number of marks strictly below the water level among all days.

Examples

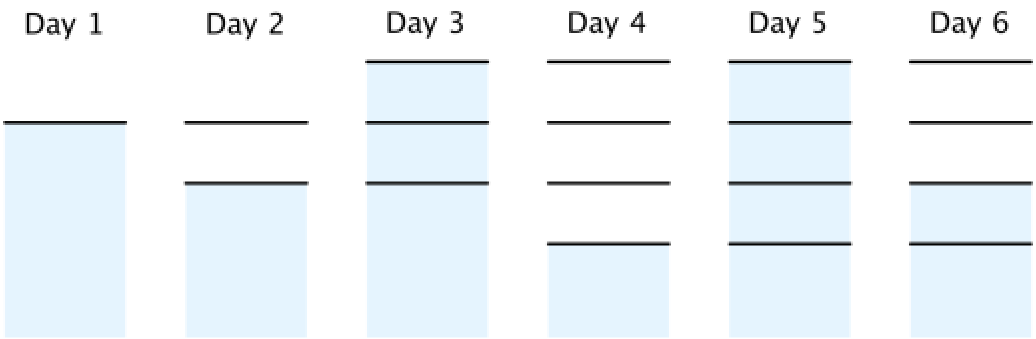
Input	Output
6 0 1 0 3 0 2	6

Input	Output
5 0 1 2 1 2	1

Input	Output
5 0 1 1 2 2	0

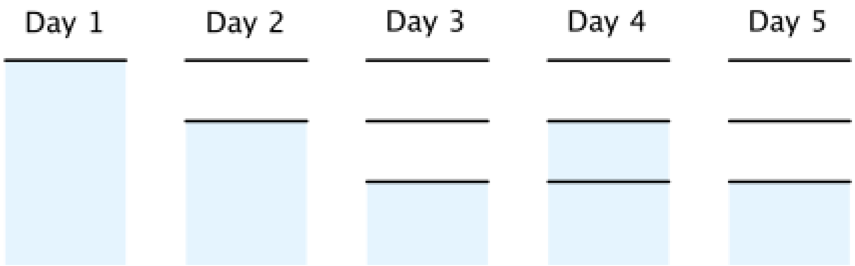
Note

In the first example, the following figure shows an optimal case.



Note that on day 3, a new mark should be created because if not, there cannot be 3 marks above water on day 4. The total number of marks underwater is $0 + 0 + 2 + 0 + 3 + 1 = 6$.

In the second example, the following figure shows an optimal case.



Problem F. The Fair Nut and Strings

Time Limit 1000 ms
Mem Limit 262144 kB

Recently, the Fair Nut has written k strings of length n, consisting of letters "a" and "b". He calculated c — the number of strings that are prefixes of at least one of the written strings. **Every string was counted only one time**.

Then, he lost his sheet with strings. He remembers that all written strings were lexicographically **not smaller** than string s and **not bigger** than string t. He is interested: what is the maximum value of c that he could get.

A string a is lexicographically smaller than a string b if and only if one of the following holds:

- a is a prefix of b, but $a \neq b$;
- in the first position where a and b differ, the string a has a letter that appears earlier in the alphabet than the corresponding letter in b.

Input

The first line contains two integers n and k ($1 \leq n \leq 5 \cdot 10^5$, $1 \leq k \leq 10^9$).

The second line contains a string s ($|s| = n$) — the string consisting of letters "a" and "b.

The third line contains a string t ($|t| = n$) — the string consisting of letters "a" and "b.

It is guaranteed that string s is lexicographically not bigger than t.

Output

Print one number — maximal value of c.

Examples

Input	Output
2 4 aa bb	6

Input	Output
3 3 aba bba	8

Input	Output
4 5 abbb baaa	8

Note

In the first example, Nut could write strings "aa", "ab", "ba", "bb". These 4 strings are prefixes of at least one of the written strings, as well as "a" and "b". Totally, 6 strings.

In the second example, Nut could write strings "aba", "baa", "bba".

In the third example, there are only two different strings that Nut could write. If both of them are written, $c = 8$.